FRANZ MICHAEL L. CAYANAN

Full-Stack & AI Application Developer | Next.js · PostgreSQL · Supabase · Flutter · Laravel · LLM, RAG & MCP Integration
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PROFESSIONAL SUMMARY

Magna Cum Laude Computer Science graduate and full-stack developer (Next.js, TypeScript, PostgreSQL, Supabase, Flutter, Laravel). Leading MedStock, a freelance multi-company inventory and invoicing system with an append-only stock ledger and Row Level Security. Experienced in LLM and MCP integration and AI-assisted development with Claude Code, a DeepSeek-based coding harness, Copilot, Cursor, and Cline, working in a spec-driven, plan–execute–verify loop. Built a multilingual RAG system that lifted Recall@4 from 79% to 91.5%, and interned at BMWare as a system architect.

EDUCATION

Bachelor of Science in Computer Science — Magna Cum Laude **Graduated September 2026**
Pampanga State University
- Relevant coursework: Software Engineering, Database Management, Artificial Intelligence, Mobile Development | Leadership: AI Lead, Technical Research and Development

PROFESSIONAL EXPERIENCE

Freelance Full-Stack Developer | MedStock (client project) **Oct 2026 – Present**
Next.js · TypeScript · PostgreSQL 16 (RLS) · Drizzle ORM · Better Auth · Vitest · GitHub Actions · Windows host + Tailscale
- Building an inventory, sales, and invoicing system for a medical supplies importer running several companies on one shared stock pool; scoped requirements with the owner and accountant and handled change requests.
- Designed a 42-table PostgreSQL schema with forced Row Level Security on 23 tables, composite foreign keys that bind child rows to their company, and a restricted app role that cannot write stock or audit tables directly.
- Enforced an append-only stock ledger: SECURITY DEFINER posting functions update the ledger and balances in one locked transaction, with signed-quantity checks and a consistency check that rebuilds every balance from the ledger.
- Built gap-free per-company invoice numbering, including linked non-sales extension invoices issued with consecutive numbers under concurrent sessions, and kept revenue reports on a sales-only view guarded by a scan test.
- Directed AI coding agents (Claude Code and a DeepSeek-based harness) with a written spec, AGENTS.md rules, and a project skill; gated each milestone with plan approval and exit tests, backed by 170 automated tests.

Technical Intern & System Architect | BMWare **Feb 2026 – Apr 2026**
- Delivered modular, cross-platform Flutter apps across multiple production projects, connected to a Laravel (PHP) backend through RESTful APIs.
- Architected the data layer (ERDs) and microservices for Adsumus Dispatch, a real-time emergency response platform with fallback state synchronization, encrypted messaging, and geolocation tracking.
- Kept payment data off the device in InsureMe, a production digital insurance app, by isolating the PayPal SDK behind a decoupled abstraction layer that separates payment state from app logic.

PROJECTS

Multilingual Research-Paper Q&A (RAG) — Sole Developer **Personal Project**
Qdrant · BAAI/bge-m3 · bge-reranker-v2-m3 · Agentic RAG · Evaluation Harness
- Raised retrieval Recall@4 from 79.3% to 91.5% (MRR 0.677 to 0.794) on 82 hand-verified questions by adding cross-encoder reranking to a multilingual Qdrant index of 32 PDFs (2,210 chunks, 4 languages); depth ablations showed gains saturate at 20 candidates.
- Prevented fabricated answers: a bounded agent loop (max 3 rounds) refuses when evidence is missing and can only cite chunks it retrieved.
- Made ingestion crash-safe (deterministic uuid5 IDs, idempotent re-ingestion) and the pipeline testable offline, with 35 tests running in CI.

Developer Workplace — Sole Developer **Personal Project · Case Study**
Next.js · React · TypeScript · Supabase (PostgreSQL, Auth, Storage, RLS) · Google Gemini API · GitHub App · dnd-kit
- Built a full-stack developer workspace (projects, Kanban, calendar, plans, notes, goals), using Supabase Auth and PostgreSQL Row Level Security to isolate each user's data; dnd-kit boards and calendar with optimistic updates and timezone-safe dates.
- Integrated a read-only GitHub App using server-side App JWTs, short-lived installation tokens, PKCE, and installation ownership checks.
- Implemented separate private and public Gemini assistants with server-verified ownership, bounded context, and Supabase-backed rate limiting; the public assistant reads only owner-published data through PostgreSQL projections/RPCs.

MCP Server Integrations (Blender, Roblox & Custom) **Developer Personal Projects**
MCP (Model Context Protocol) · Blender · Roblox · Custom MCP Server · LLM Agents
- Connected LLM agents to Blender, Roblox, and a custom-built MCP server, exposing application actions as tools AI clients can call.

AI-Powered Maintenance Simulator — Project Lead & System Architect **Thesis Project, 2025 – 2026**
Unity · C# · YOLOv11 · Unity Sentis · Firebase Firestore
- Led the thesis and designed the architecture (DFDs, ERDs) for a real-time car part recognition app running YOLOv11 on-device through Unity Sentis, with a Firestore pipeline for real-time sync.

TECHNICAL SKILLS

Languages & Frameworks: TypeScript, JavaScript, Dart, C#, PHP, Java, C++, SQL; Next.js, React, Flutter, Laravel, .NET, Node.js, Unity
AI & Automation: RAG, vector databases (Qdrant), embeddings, cross-encoder reranking, retrieval evaluation (Recall@k, MRR), Google Gemini API, Ollama (local LLMs), OpenAI/Anthropic APIs, MCP servers, prompt engineering, n8n, agentic workflows, YOLOv11, Unity Sentis
AI Dev Workflow: Spec-driven planner + coding-agent workflow (Claude Code, DeepSeek-based harness, Codex, Copilot, Cursor, Cline), AGENTS.md/CLAUDE.md rules, project skills, milestone plan-approval gates
Databases & Backend: PostgreSQL (RLS, triggers, PL/pgSQL), Drizzle ORM, Supabase, Better Auth, Qdrant, MySQL, MS SQL Server, Firestore, REST APIs
Tools & Practices: Git, GitHub Actions, Vitest, Docker, Vercel, JIRA, Agile/Scrum, system design (ERD, DFD), UI/UX